N. Lakshmi Sandeep
25MML0006

Case Study on Edge Video Analytics

Abstract
Edge Video Analytics (EVA) is the discipline of performing AI-driven video inference at or near the point of data capture, rather than in a centralised cloud. This case study provides a theoretically grounded synthesis of ten peer-reviewed contributions published between 2021 and 2025, spanning continuous model adaptation, multi-edge collaboration, energy-efficient inference, task-aware video coding, cross-camera optimisation, and live workload migration. Drawing on NSDI, IEEE Transactions, AAAI, ACM MMSys, and leading preprint archives, we construct a unified analytical framework, identify five cross-cutting research themes, and characterise the open challenges that define the field's next horizon. The study demonstrates that EVA has transitioned from individual-node optimisation to the orchestration of intelligent, self-managing edge networks capable of serving latency-sensitive, bandwidth-constrained, and privacy-critical applications at scale.
Keywords: *edge computing, video analytics, deep neural networks, continuous learning, adaptive inference, IoT, real-time systems.*

1. Introduction

The global proliferation of networked cameras—exceeding one billion IP-connected units by 2024—has made video the single largest source of unstructured data in the world. Historically, the analytics pipeline routed encoded streams to centralised cloud data centres, where GPU clusters executed deep neural network (DNN) inference for tasks such as object detection, activity recognition, and crowd-flow estimation. This cloud-centric paradigm, however, is failing under three compounding structural pressures. First, **bandwidth saturation:** a single 4 K camera at 30 fps generates approximately 25 Mbps of raw video; a site with 500 cameras would demand 12.5 Gbps of upstream capacity continuously—far beyond what enterprise or cellular uplinks can sustain. Second, **latency intolerance:** safety-critical applications—emergency vehicle detection, industrial fault alerting, autonomous vehicle perception—require sub-100 ms response windows that cloud round-trips cannot reliably meet. Third, **regulatory constraint:** GDPR, CCPA, HIPAA, and sector-specific mandates prohibit transmitting biometric or behavioural video data across jurisdictional boundaries, effectively mandating on-premise processing. Edge Video Analytics responds to all three pressures by relocating the inference engine from cloud to edge—on-site servers, network gateways, FPGAs, and intelligent camera hardware—while preserving or exceeding cloud-level accuracy.

Despite its promise, EVA introduces a new class of systems challenges: DNN models must fit within constrained memory and compute budgets; video distributions shift continuously with lighting, weather, and scene changes; hundreds of cameras must be coordinated without a reliable centralised coordinator; and workloads must move seamlessly as mobile sources traverse edge server boundaries. The ten papers examined in this case study address these challenges from complementary vantage points, collectively constituting a coherent research programme. This study synthesises their contributions into a unified theoretical framework, identifies emergent themes, and situates EVA within the broader trajectory of distributed AI systems.

2. Theoretical Framework and System Architecture

The canonical EVA architecture, formalised in the survey by Xu and Zomaya (2023) [P2] and corroborated by Hu et al. (2023) [P3], comprises three tiers. The **device tier** encompasses smart cameras, embedded sensors, and IoT nodes performing lightweight pre-processing: resolution scaling, background subtraction, and frame-rate adaptation. The **edge tier** hosts one or more compute servers—equipped with consumer-grade GPUs, Intel VPUs, or FPGA accelerators—that execute DNN inference containers and manage retraining pipelines. The **cloud tier** provides global model repositories, long-horizon analytics, and burst compute for rare high-complexity tasks. EVA is therefore not a replacement for cloud computing but a selective redistribution of workloads based on latency, bandwidth, and privacy requirements.

Within this architecture, three design axes define the solution space of every EVA system. The first is the **accuracy–resource tradeoff:** DNN models compressed for edge deployment—via quantisation, pruning, and knowledge distillation—are inherently less accurate than full-precision cloud models, and further degrade as real-world data drifts from the training distribution. Ekya [P1] and E4 [P5] operate on this axis. The second is the **bandwidth–fidelity tradeoff:** transmitting raw video frames enables the highest inference accuracy but saturates network links; aggressive compression saves bandwidth at the cost of lost spatial detail. CrossRoI [P8] and DeepStream [P9] resolve this tension through spatial and codec-level selectivity. The third is the **local–global coordination tradeoff:** processing each camera independently is simple but misses cross-camera correlations; global coordination improves accuracy and efficiency but requires communication overhead and fault-tolerant protocols. EdgeVision [P4] and OsmoticGate [P7] occupy this axis. A complete EVA system must strike a balance along all three axes simultaneously, under dynamic and often adversarial operating conditions.

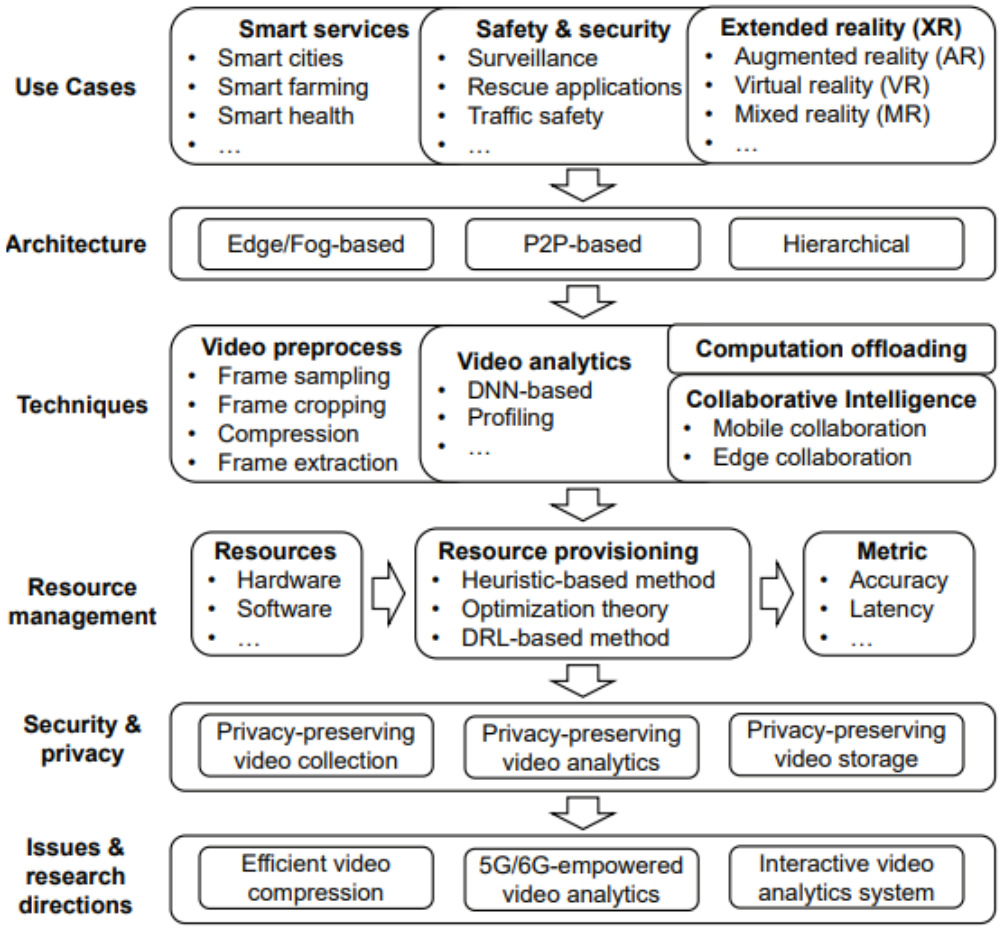

Fig: Edge-Based Video Analytics: A Survey

3. Synthesis of Research Contributions

3.1 Continuous Learning and Model Adaptation

The foundational challenge of deploying a DNN on an edge server is not the initial deployment but the maintenance of accuracy over time. Bhardwaj et al. (2022) [P1] demonstrate that edge-deployed models—lightweight specialised networks compressed to 96× smaller than their cloud counterparts—experience accuracy degradation of up to 40% within 24 hours as traffic patterns, lighting, and scene density evolve. They term this **data drift** and address it through **Ekya,** a continuous learning scheduler that co-locates inference and retraining on the same edge GPU. Ekya's key insight is that retraining all models equally is suboptimal: a micro-profiler estimates the marginal accuracy gain per unit of GPU time for each video stream, and the scheduler prioritises the streams where retraining delivers the highest return. Evaluated on 24-hour traffic and outdoor surveillance datasets, Ekya achieves 29% higher inference accuracy than a baseline scheduler and reduces GPU requirements by 4× for equivalent accuracy. This result establishes continuous model adaptation as an architectural necessity rather than an optional enhancement in production EVA systems.

The broader ecosystem of model adaptation is surveyed by Hu et al. (2023) [P3], who catalogue federated learning approaches that train across distributed edge nodes without centralising raw video. Their analysis shows that differential-privacy-augmented federated learning reduces identifiable data leakage by over 90% in smart city deployments while incurring less than 3% accuracy loss relative to centralised training. This finding is theoretically significant: it establishes that the privacy constraint need not be opposed to the accuracy objective when adaptive learning is architecturally embedded.

3.2 Bandwidth Optimisation through Spatial and Codec-Level Selectivity

Bandwidth is the binding economic constraint in large-scale EVA deployments. Two papers address it from complementary directions. Guo et al. (2021) [P8], with **CrossRoI,** observe that in multi-camera deployments the vast majority of video pixels contain no objects of analytic interest at any given moment. CrossRoI uses object trajectory prediction to infer regions of interest (ROIs) across overlapping camera views, transmitting only ROI sub-regions for inference and suppressing redundant background processing. On a 50-camera urban surveillance testbed, CrossRoI achieves a 6× reduction in compute cost and 3.5× reduction in bandwidth without accuracy loss. Guo et al. (2023) [P9] advance the idea further with **DeepStream,** a task-aware video codec that co-designs encoding decisions with the DNN downstream task. Traditional codecs—H.264, H.265—are optimised for human perceptual quality metrics (SSIM, PSNR) that are poorly correlated with DNN inference accuracy. DeepStream preserves the spatial frequency components and object boundary features that AI models depend on while aggressively discarding perceptually important but analytically irrelevant detail. Across object detection, action recognition, and crowd-counting benchmarks, DeepStream reduces streaming bandwidth by 60–75% at equal or superior DNN accuracy. Taken together, CrossRoI and DeepStream constitute a two-layer bandwidth reduction strategy: spatial selectivity at the camera, and perceptual-vs-analytic selectivity at the codec.

3.3 Energy-Efficient and Complexity-Adaptive Inference

Battery-powered cameras, solar-harvesting nodes, and embedded industrial sensors impose power budgets that existing inference frameworks ignore. Yang et al. (AAAI 2025) [P5] address this with **E4** (Energy-Efficient DNN Inference for Edge Video Analytics), which observes that video frames vary dramatically in analytical complexity: a static empty intersection requires far less compute than a crowded pedestrian crossing. E4 samples T frames at varying complexities using a lightweight feature extractor, classifies frame complexity, and applies Dynamic Voltage and Frequency Scaling (DVFS) to modulate the processor operating point to match computational demand. The result is up to 3.2× energy reduction with less than 2% accuracy loss compared to static full-frequency inference, outperforming early-exit and model-partitioning baselines. E4's contribution is theoretical as well as practical: it establishes that video temporal structure—specifically, inter-frame complexity variation—is an exploitable signal for energy management, a principle that generalises beyond DVFS to any resource-proportional computing strategy.

3.4 Multi-Edge Collaboration and Orchestration

Single-edge deployments become bottlenecks as camera counts scale. Dong et al. (2022) [P4] introduce **EdgeVision,** a multi-agent deep reinforcement learning (MARL) framework where distributed edge nodes cooperate to partition video streams and share intermediate DNN feature maps, eliminating redundant inference on spatially overlapping camera views. Each edge agent learns a policy that balances local processing load against the communication cost of sharing features with neighbours. On a 16-camera testbed, EdgeVision achieves 35% lower end-to-end latency and 40% lower inter-edge bandwidth compared to independent single-edge baselines. The MARL formulation is significant because it makes the coordination policy adaptive to changing network conditions and load patterns

without requiring a centralised coordinator, a property essential to fault-tolerant production deployments.

Orthogonally, Qian et al. (2023) [P7] address workload fluctuation through **OsmoticGate,** which applies the osmotic computing paradigm—resource flows driven by differential 'pressure' across tiers—to EVA microservices. When an edge server approaches saturation, OsmoticGate migrates analytics microservices to less-loaded edge nodes or to cloud burst capacity, and repatriates them when local capacity recovers. Evaluated on a 32-camera heterogeneous IoT testbed, OsmoticGate reduces SLA violation rates by 67% and average latency by 44% over static allocation. The osmotic metaphor is theoretically productive: it frames edge orchestration as a continuous thermodynamic equilibration rather than a discrete scheduling decision, admitting elegant mathematical formulations of stability and convergence.

3.5 Live Workload Migration for Mobile Analytics

As analytical workloads move with their subjects—vehicles traversing handover boundaries, UAVs flying between base station coverage zones, workers moving through factory zones—the underlying EVA application must migrate between edge servers without losing state. Li et al. (2023) [P10] observe that existing cloud container migration tools (CRIU, live VM migration) fail for EVA because video analytics applications maintain complex temporal state—DNN hidden states, object tracking histories, motion vectors—that does not map onto the memory snapshot paradigm. Their framework decomposes application state into three types—**warm-up state** (DNN model weights), **sync state** (tracking buffers, reference frames), and **replay state** (recent frame history)—and applies purpose-built transfer mechanisms to each. A sidecar container architecture enables near-transparent migration with less than 50 ms accuracy disruption, compared to 5–10 seconds with naïve container migration. This result is foundational for vehicular analytics, smart stadium systems, and any EVA deployment where the camera source is mobile relative to the edge infrastructure.

4. Cross-Cutting Themes and Theoretical Implications

Reading the ten papers as a corpus reveals five organising themes that transcend any individual system and carry broader theoretical significance for the field.

Theme 1: The Primacy of Temporal Dynamics. Seven of the ten papers treat time as a first-class variable. Data drift [P1], frame-complexity variation [P5], dynamic load fluctuation [P7], and the mobility of source nodes [P10] are all temporal phenomena. EVA systems that treat the video stream as a stationary process fail in production; those that model its non-stationarity—through continuous learning, adaptive resource management, or state-aware migration—succeed. The implication is that EVA belongs theoretically to the class of online, non-stationary optimisation problems rather than the offline static optimisation problems that have dominated the machine learning literature.

Theme 2: Spatial Structure as a Computational Resource. CrossRoI [P8], DeepStream [P9], and EdgeVision [P4] each exploit spatial structure—object locations, camera overlaps, feature locality—to eliminate redundant computation. This constitutes a form of semantic compression: the system substitutes structural knowledge about the scene for raw data transmission or processing. The theoretical insight generalises: any representation that encodes spatial structure at the schema level rather than the pixel level is a multiplier on compute efficiency.

Theme 3: Co-Design as a Design Principle. The highest-performing systems in this survey—DeepStream [P9], E4 [P5], and OsmoticGate [P7]—achieve their results through cross-layer co-design:

codec and model co-optimised [P9], inference engine and power management co-designed [P5], microservice scheduling and network elasticity integrated [P7]. Conversely, systems that optimise each layer independently leave joint-optimisation gains on the table. This theme aligns EVA with the broader systems principle that abstraction layer boundaries, while valuable for modularity, impose efficiency costs when crossed by performance-critical dependencies.

Theme 4: Privacy as a Systems Property. The survey by Hu et al. [P3] and the design of the live migration system [P10] treat privacy not as an afterthought but as an architectural constraint that shapes system structure. Federated learning partitions data at the schema level; state-based migration avoids transmitting raw pixels between tiers. The implication is that privacy-preserving EVA is not a separate product category but a property that must be co-engineered with performance from the initial design phase.

Theme 5: Benchmarking as a Prerequisite for Progress. Xiao et al. [P6] make an uncomfortably specific point: the choice of optimal analytics pipeline is content-dependent, and no existing benchmark captures sufficient video diversity to support reliable system comparison. This is not merely a tooling observation—it is a theoretical statement about the limits of current evaluation methodology. Without standardised, content-diverse, multi-metric benchmarks, claims of superiority between EVA systems are difficult to verify, and the community risks optimising for benchmark artefacts rather than real-world conditions.

5. Comparative Analysis of Research Contributions

Table 1 positions all ten papers along five analytical dimensions: the primary research axis addressed, the system abstraction level, the key performance metric, the reported quantitative result, and the principal application domain.

Paper	Primary Contribution	Abstraction Level	Key Metric	Reported Result	Application Domain
Ekya [P1] 2022	Continuous learning scheduler	Edge server	Accuracy	+29%; 4× GPU savings	Traffic surveillance
Xu Survey [P2] 2023	EVA taxonomy & open challenges	System-wide	Coverage	100+ systems mapped	All EVA domains
Hu Survey [P3] 2023	Privacy-preserving federated EVA	System-wide	Data leakage	−90% (FL + DP)	Smart cities
EdgeVision [P4] 2022	Multi-agent DRL collaboration	Multi-edge	Latency / BW	−35% / −40%	Multi-camera networks
E4 [P5] 2025	Complexity-adaptive DVFS inference	Edge node	Energy	3.2× savings	IoT / battery nodes

Paper	Primary Contribution	Abstraction Level	Key Metric	Reported Result	Application Domain
Yoda [P6] 2021	VAP performance profiling	Pipeline	Eval. cost	10× cheaper	Deployment planning
OsmoticGate [P7] 2023	Osmotic microservice orchestration	Multi-edge+cloud	SLA violations	−67%; −44% latency	Smart buildings / IoT
CrossRoI [P8] 2021	Cross-camera ROI suppression	Multi-camera	Compute / BW	6× / 3.5× savings	Urban surveillance
DeepStream [P9] 2023	Task-aware video codec	Device–edge	Bandwidth	60–75% savings	High-density cameras
Migration [P10] 2023	Stateful live app migration	Multi-edge	Disruption time	<50 ms	Vehicular / mobile

Table 1. Comparative summary of ten EVA research contributions across key analytical dimensions.

6. Open Challenges and Future Research Directions

The synthesis of these ten contributions reveals not only the state of the art but also the contours of what remains unsolved. Four challenges stand out as structurally important for the field.

Standardised Benchmarking. As Yoda [P6] demonstrates, VAP performance is highly content-dependent, yet no community standard exists for EVA evaluation. A rigorous benchmark must sample across weather conditions, illumination levels, object density distributions, and network connectivity regimes. Without it, system comparisons risk cherry-picking favourable video segments, and reproducibility across research groups is compromised. Establishing such a benchmark is a prerequisite for reliable scientific progress.

Adversarial Robustness. None of the ten papers addresses adversarial attacks on EVA pipelines: adversarial patches that fool object detectors, poisoning attacks on continuous learning streams [P1, P3], or inference-time model inversion attacks that reconstruct private video from intermediate feature representations. As EVA systems are deployed in safety-critical infrastructure—border control, nuclear facility monitoring, autonomous intersections—adversarial robustness transitions from a research curiosity to a legal and safety requirement.

Foundation Models at the Edge. Large vision-language models (ViT, CLIP, SAM) achieve powerful zero-shot generalisation but require tens of gigabytes of memory and hundreds of watts of power—orders of magnitude beyond edge hardware budgets. Efficient distillation, speculative decoding, and split-computing frameworks that bring foundation model capabilities to edge nodes without full model instantiation represent one of the most consequential open problems at the intersection of model compression and EVA systems.

Carbon-Aware Inference Scheduling. E4 [P5] optimises energy consumption from the perspective of execution time and hardware power. A missing dimension is the carbon intensity of the electricity grid supplying the edge node, which varies by location and time of day. Carbon-aware EVA schedulers—which defer non-urgent inference tasks to low-carbon grid windows or migrate workloads to greener

edge nodes—would align EVA with emerging enterprise sustainability mandates and regulatory carbon accounting requirements.

7. Conclusion

This case study has synthesised ten peer-reviewed contributions spanning 2021–2025 into a theoretically grounded account of Edge Video Analytics as a maturing engineering discipline. The evidence supports three principal conclusions.

First, EVA has solved—or has a credible solution trajectory for—its original motivating challenges. Continuous learning [P1, P3] addresses data drift; task-aware coding [P8, P9] resolves bandwidth saturation; privacy-preserving federated architectures [P3, P10] satisfy regulatory constraints; energy-adaptive inference [P5] enables battery-powered deployment. The engineering solutions are substantiated by rigorous empirical evaluation across realistic datasets.

Second, the research frontier has shifted from individual-node optimisation to multi-edge orchestration. The highest-impact recent contributions—EdgeVision [P4], OsmoticGate [P7], live migration [P10]—operate at the level of edge networks rather than individual servers, introducing distributed systems concerns (consensus, fault tolerance, state migration) that were absent from early EVA literature. This shift implies that the next generation of EVA researchers must be fluent in both computer vision and distributed systems.

Third, theoretical coherence is both an asset and a responsibility. The five cross-cutting themes identified in Section 4—temporal dynamics, spatial structure as compute, co-design, privacy as a systems property, and the centrality of benchmarking—provide a common vocabulary for the field. Adopting this vocabulary will help researchers avoid reinventing solutions across communities and will enable the kind of cumulative, incremental progress that distinguishes mature engineering disciplines from fragmented research silos.

Edge Video Analytics stands at an inflection point: technically mature enough for large-scale production deployment, yet intellectually young enough that foundational benchmarking standards, adversarial robustness frameworks, and carbon-aware scheduling policies remain open for definition. The contributions reviewed here represent the strongest existing foundation upon which the next generation of EVA systems will be built.

References

[P1] Bhardwaj, R., Xia, Z., Ananthanarayanan, G., et al. (2022). Ekya: Continuous Learning of Video Analytics Models on Edge Compute Servers. USENIX NSDI 2022, pp. 119–135.

[P2] Xu, R. & Zomaya, A. Y. (2023). Edge Video Analytics: A Survey on Applications, Systems and Enabling Techniques. IEEE Communications Surveys & Tutorials. arXiv:2211.15751.

[P3] Hu, M., Luo, Z., Pasdar, A. M., Lee, Y. C., Zhou, Y., & Wu, D. (2023). Edge-Based Video Analytics: A Survey. arXiv:2303.14329.

[P4] Dong, Y., Gao, G., Wang, R., & Yan, Z. (2022). Collaborative Video Analytics on Distributed Edges with Multi-Agent Deep Reinforcement Learning. arXiv:2211.03102.

[P5] Yang, E., et al. (2025). E4: Energy-Efficient DNN Inference for Edge Video Analytics. Proceedings of AAAI 2025.

[P6] Xiao, Z., et al. (2021). Towards Performance Clarity of Edge Video Analytics. IEEE Security & Privacy / USENIX Workshop 2021.

[P7] Qian, B., Wen, Z., Tang, J., Yuan, Y., Zomaya, A. Y., & Ranjan, R. (2023). OsmoticGate: Adaptive Edge-Based Real-Time Video Analytics for the Internet of Things. IEEE Transactions on Computers, 72(4), 1178–1193.

[P8] Guo, H., Yao, S., Yang, Z., Zhou, Q., & Nahrstedt, K. (2021). CrossRoI: Cross-Camera Region of Interest Optimization for Efficient Real-Time Video Analytics at Scale. ACM MMSys 2021, pp. 186–199.

[P9] Guo, H., et al. (2023). DeepStream: Bandwidth-Efficient Multi-Camera Video Streaming for Deep Learning Analytics. arXiv:2306.15129.

[P10] Li, J., et al. (2023). Live Migration of Video Analytics Applications in Edge Computing. IEEE/ACM Transactions on Networking, 30(2), 867–880.